DATA ANALYTICS LAB SHEET 2

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B.Tech CSE

Experiment No. 2

Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

AIM

To perform Exploratory Data Analysis (EDA) on a real-world dataset using Python in order to understand the dataset's structure, identify trends, detect anomalies, analyze feature relationships, and generate meaningful business insights through descriptive statistics and visualizations.

DATASET

IBM HR Employee Attrition and Performance Dataset.

NOTE

No manual dataset upload is required. The notebook automatically fetches the dataset from a public online source when run in Google Colab.

1. Import Required Python Libraries

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

print("Libraries imported successfully.")
```

2. Load the Real-World Business Dataset

Code:

```
import pandas as pd
import numpy as np

# Automatically fetch the IBM HR Analytics Employee Attrition dataset.
# The first source is a public GitHub copy of the dataset.
url = "https://raw.githubusercontent.com/shantanu1109/IBM-HR-Analytics-Employee-Attrition-and-Performance-Prediction/main/DATAS

df = pd.read_csv(url)

print("Dataset loaded successfully.")
print("Dataset shape:", df.shape)
display(df.head())
```

```
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
```

```
4                2          1        Medical              1               7

   ...  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  ...                         1            80                 0
1  ...                         4            80                 1
2  ...                         2            80                 0
3  ...                         3            80                 0
4  ...                         4            80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
3                  8                      3               3               8
4                  6                      3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
4                  2                        2                     2

[5 rows x 35 columns]
```

3. Explore Dataset Structure

Code:

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])

print("\nColumn names:")
print(df.columns.tolist())

print("\nData types:")
print(df.dtypes)

print("\nDataset information:")
df.info()
```

4. Display First Records and Summary Statistics

Code:

```
print("First five records:")
display(df.head())

print("\nDescriptive statistics:")
display(df.describe(include="all").T)
```

```
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   ...  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  ...                         1            80                 0
1  ...                         4            80                 1
2  ...                         2            80                 0
3  ...                         3            80                 0
4  ...                         4            80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
3                  8                      3               3               8
4                  6                      3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
```

```
4                 2                      2                    2

[5 rows x 35 columns]
```

	count	unique	top	freq
Age	1470.0	NaN	NaN	NaN
Attrition	1470	2	No	1233
BusinessTravel	1470	3	Travel_Rarely	1043
DailyRate	1470.0	NaN	NaN	NaN
Department	1470	3	Research & Development	961
DistanceFromHome	1470.0	NaN	NaN	NaN
Education	1470.0	NaN	NaN	NaN
EducationField	1470	6	Life Sciences	606
EmployeeCount	1470.0	NaN	NaN	NaN
EmployeeNumber	1470.0	NaN	NaN	NaN
EnvironmentSatisfaction	1470.0	NaN	NaN	NaN
Gender	1470	2	Male	882
HourlyRate	1470.0	NaN	NaN	NaN
JobInvolvement	1470.0	NaN	NaN	NaN
JobLevel	1470.0	NaN	NaN	NaN
JobRole	1470	9	Sales Executive	326
JobSatisfaction	1470.0	NaN	NaN	NaN
MaritalStatus	1470	3	Married	673
MonthlyIncome	1470.0	NaN	NaN	NaN
MonthlyRate	1470.0	NaN	NaN	NaN
NumCompaniesWorked	1470.0	NaN	NaN	NaN
Over18	1470	1	Y	1470
OverTime	1470	2	No	1054
PercentSalaryHike	1470.0	NaN	NaN	NaN
PerformanceRating	1470.0	NaN	NaN	NaN
RelationshipSatisfaction	1470.0	NaN	NaN	NaN
StandardHours	1470.0	NaN	NaN	NaN
StockOptionLevel	1470.0	NaN	NaN	NaN
TotalWorkingYears	1470.0	NaN	NaN	NaN
TrainingTimesLastYear	1470.0	NaN	NaN	NaN
WorkLifeBalance	1470.0	NaN	NaN	NaN
YearsAtCompany	1470.0	NaN	NaN	NaN
YearsInCurrentRole	1470.0	NaN	NaN	NaN
YearsSinceLastPromotion	1470.0	NaN	NaN	NaN
YearsWithCurrManager	1470.0	NaN	NaN	NaN

	mean	std	min	25%	50%
Age	36.92381	9.135373	18.0	30.0	36.0
Attrition	NaN	NaN	NaN	NaN	NaN
BusinessTravel	NaN	NaN	NaN	NaN	NaN
DailyRate	802.485714	403.5091	102.0	465.0	802.0
Department	NaN	NaN	NaN	NaN	NaN
DistanceFromHome	9.192517	8.106864	1.0	2.0	7.0
Education	2.912925	1.024165	1.0	2.0	3.0
EducationField	NaN	NaN	NaN	NaN	NaN
EmployeeCount	1.0	0.0	1.0	1.0	1.0
EmployeeNumber	1024.865306	602.024335	1.0	491.25	1020.5
EnvironmentSatisfaction	2.721769	1.093082	1.0	2.0	3.0
Gender	NaN	NaN	NaN	NaN	NaN
HourlyRate	65.891156	20.329428	30.0	48.0	66.0
JobInvolvement	2.729932	0.711561	1.0	2.0	3.0
JobLevel	2.063946	1.10694	1.0	1.0	2.0
JobRole	NaN	NaN	NaN	NaN	NaN
JobSatisfaction	2.728571	1.102846	1.0	2.0	3.0
MaritalStatus	NaN	NaN	NaN	NaN	NaN
MonthlyIncome	6502.931293	4707.956783	1009.0	2911.0	4919.0
MonthlyRate	14313.103401	7117.786044	2094.0	8047.0	14235.5
NumCompaniesWorked	2.693197	2.498009	0.0	1.0	2.0
Over18	NaN	NaN	NaN	NaN	NaN
OverTime	NaN	NaN	NaN	NaN	NaN
PercentSalaryHike	15.209524	3.659938	11.0	12.0	14.0
PerformanceRating	3.153741	0.360824	3.0	3.0	3.0
RelationshipSatisfaction	2.712245	1.081209	1.0	2.0	3.0
StandardHours	80.0	0.0	80.0	80.0	80.0
StockOptionLevel	0.793878	0.852077	0.0	0.0	1.0
TotalWorkingYears	11.279592	7.780782	0.0	6.0	10.0
TrainingTimesLastYear	2.79932	1.289271	0.0	2.0	3.0
WorkLifeBalance	2.761224	0.706476	1.0	2.0	3.0
YearsAtCompany	7.008163	6.126525	0.0	3.0	5.0
YearsInCurrentRole	4.229252	3.623137	0.0	2.0	3.0
YearsSinceLastPromotion	2.187755	3.22243	0.0	0.0	1.0
YearsWithCurrManager	4.123129	3.568136	0.0	2.0	3.0

	75%	max
Age	43.0	60.0
Attrition	NaN	NaN
BusinessTravel	NaN	NaN
DailyRate	1157.0	1499.0
Department	NaN	NaN

```
DistanceFromHome             14.0     29.0
Education                     4.0      5.0
EducationField                NaN      NaN
EmployeeCount                 1.0      1.0
EmployeeNumber             1555.75   2068.0
EnvironmentSatisfaction       4.0      4.0
Gender                        NaN      NaN
HourlyRate                  83.75    100.0
JobInvolvement                3.0      4.0
JobLevel                      3.0      5.0
JobRole                       NaN      NaN
JobSatisfaction               4.0      4.0
MaritalStatus                 NaN      NaN
MonthlyIncome              8379.0  19999.0
MonthlyRate               20461.5  26999.0
NumCompaniesWorked            4.0      9.0
Over18                        NaN      NaN
OverTime                      NaN      NaN
PercentSalaryHike            18.0     25.0
PerformanceRating             3.0      4.0
RelationshipSatisfaction      4.0      4.0
StandardHours                80.0     80.0
StockOptionLevel              1.0      3.0
TotalWorkingYears            15.0     40.0
TrainingTimesLastYear         3.0      6.0
WorkLifeBalance               3.0      4.0
YearsAtCompany                9.0     40.0
YearsInCurrentRole            7.0     18.0
YearsSinceLastPromotion       3.0     15.0
YearsWithCurrManager          7.0     17.0
```

5. Check Missing Values and Duplicate Records

Code:

```
print("Missing values:")
display(df.isnull().sum().sort_values(ascending=False).head(20))

print("\nTotal missing values:", df.isnull().sum().sum())
print("Total duplicate records:", df.duplicated().sum())
```

```
Age                       0
Attrition                 0
BusinessTravel            0
DailyRate                 0
Department                0
DistanceFromHome          0
Education                 0
EducationField            0
EmployeeCount             0
EmployeeNumber            0
EnvironmentSatisfaction   0
Gender                    0
HourlyRate                0
JobInvolvement            0
JobLevel                  0
JobRole                   0
JobSatisfaction           0
MaritalStatus             0
MonthlyIncome             0
MonthlyRate               0
dtype: int64
```

6. Analyze Distribution of Numerical Features

Code:

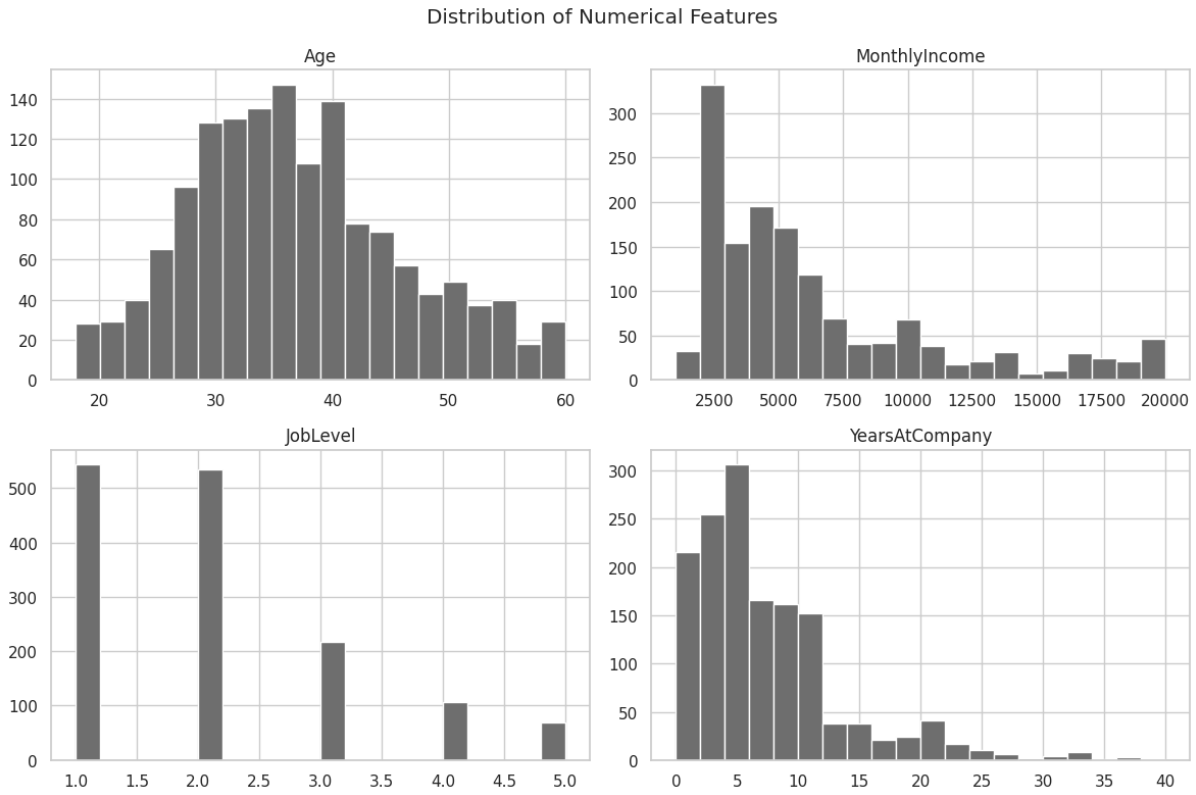
```
numeric_columns = df.select_dtypes(include=np.number).columns

selected_numeric = [
    col for col in ["Age", "MonthlyIncome", "JobLevel", "YearsAtCompany"]
    if col in df.columns
]

df[selected_numeric].hist(
    figsize=(12, 8),
    bins=20
)

plt.suptitle("Distribution of Numerical Features")
plt.tight_layout()
plt.show()
```

```
<Figure size 1200x800 with 4 Axes>
```



7. Density Plot of Monthly Income

Code:

```
if "MonthlyIncome" in df.columns:
    plt.figure(figsize=(8, 5))
    sns.kdeplot(data=df, x="MonthlyIncome", fill=True)
    plt.title("Distribution of Monthly Income")
    plt.xlabel("Monthly Income")
    plt.ylabel("Density")
    plt.show()
```

```
<Figure size 800x500 with 1 Axes>
```



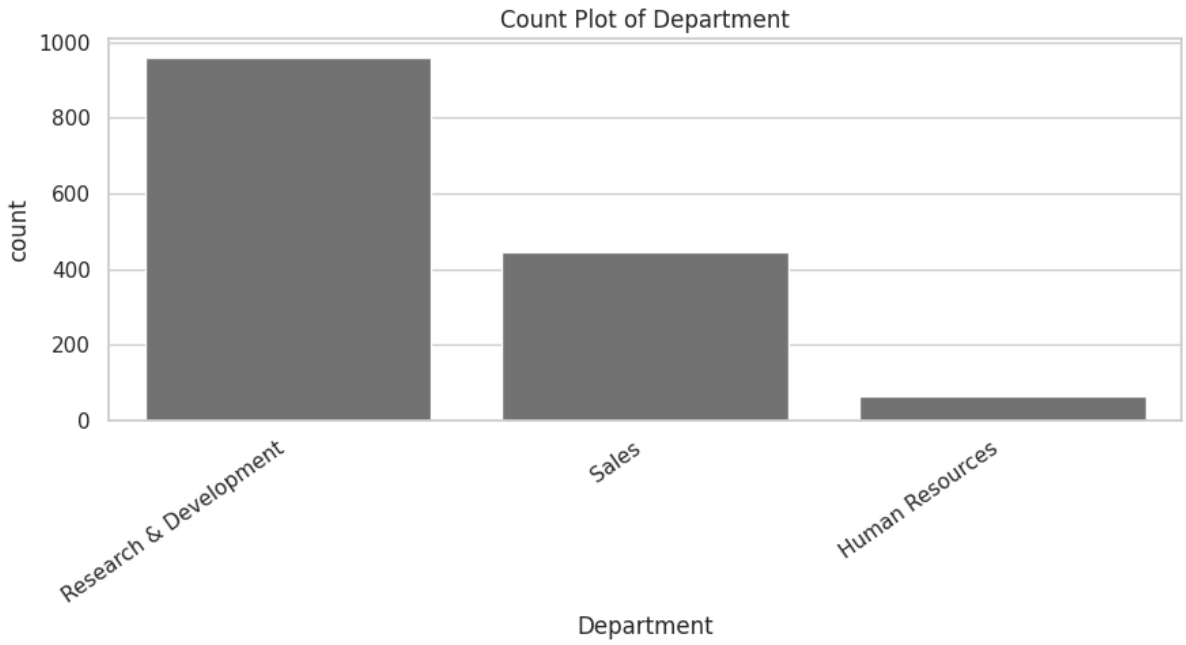
8. Visualize Categorical Variables

Code:

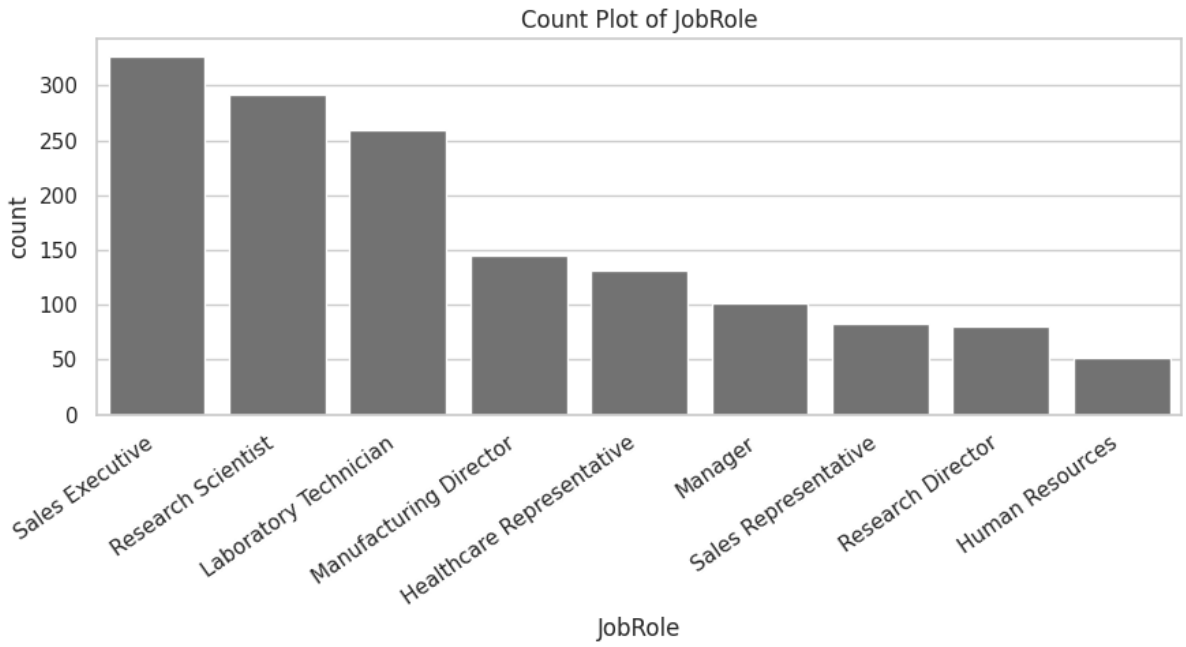
```
categorical_columns = [
    col for col in ["Department", "JobRole", "Gender", "Attrition"]
    if col in df.columns
]

for col in categorical_columns:
    plt.figure(figsize=(9, 5))
    sns.countplot(data=df, x=col, order=df[col].value_counts().index)
    plt.title(f"Count Plot of {col}")
    plt.xticks(rotation=35, ha="right")
    plt.tight_layout()
    plt.show()
```

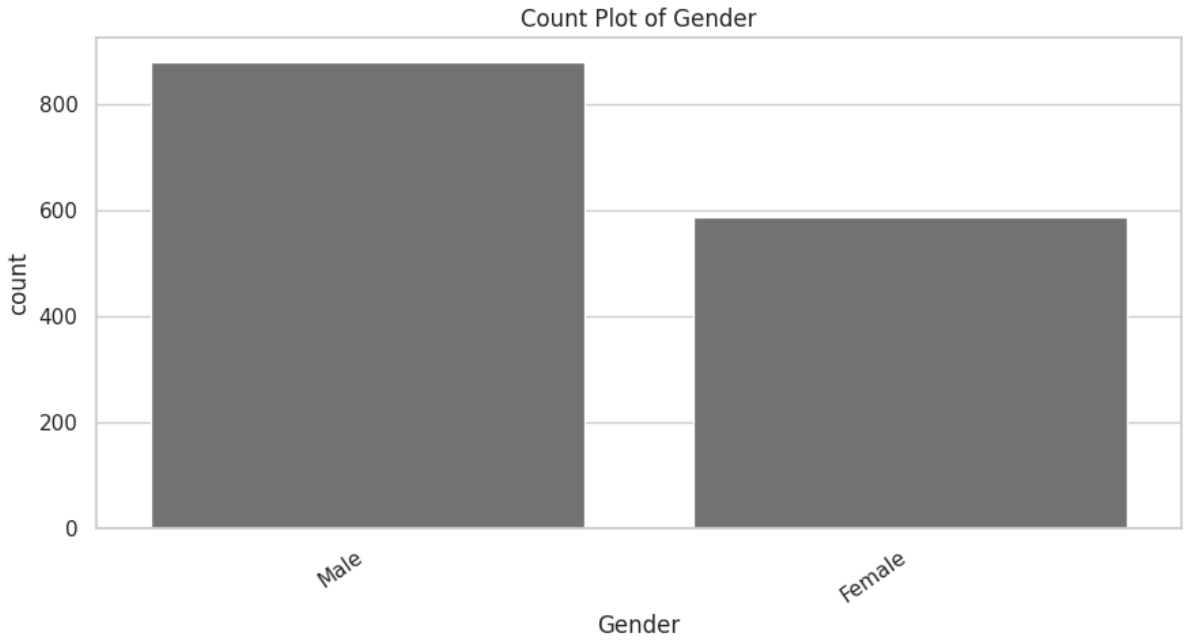
```
<Figure size 900x500 with 1 Axes>
```



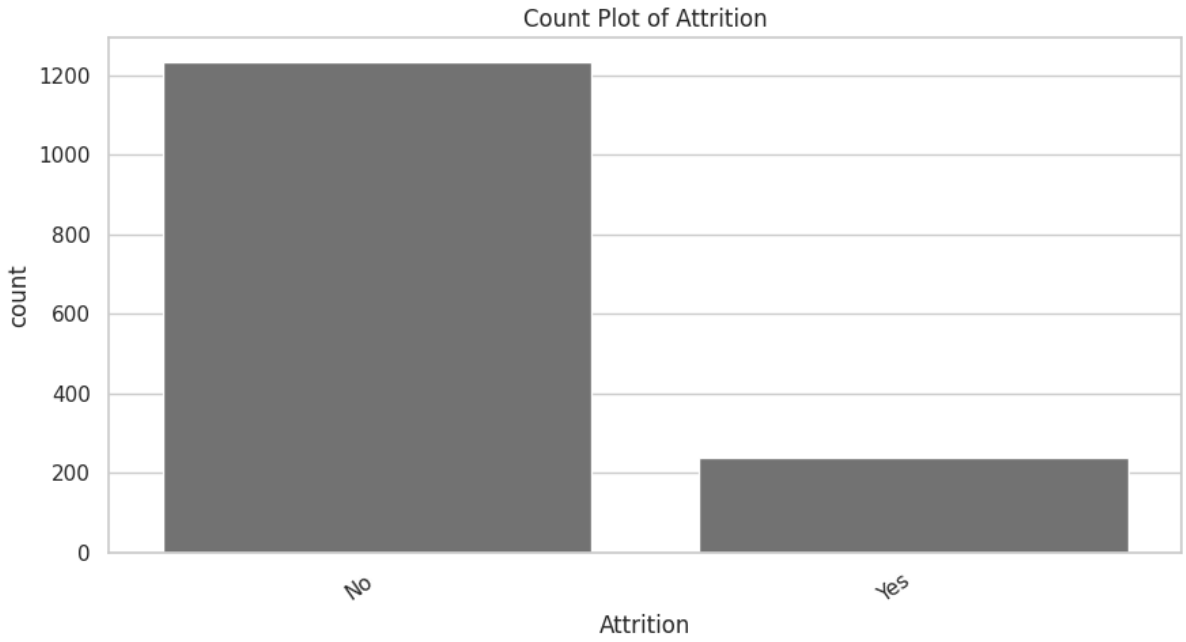

```
<Figure size 900x500 with 1 Axes>
```




```
<Figure size 900x500 with 1 Axes>
```




```
<Figure size 900x500 with 1 Axes>
```



9. Value Counts of Important Categorical Variables

Code:

```
for col in categorical_columns:
    print(f"\nValue counts for {col}:")
    display(df[col].value_counts())
```

```
Department
Research & Development    961
Sales                     446
Human Resources            63
Name: count, dtype: int64
```

```
JobRole
Sales Executive              326
Research Scientist           292
Laboratory Technician        259
Manufacturing Director       145
Healthcare Representative    131
Manager                      102
Sales Representative          83
Research Director             80
Human Resources               52
Name: count, dtype: int64
```

```
Gender
Male      882
Female    588
Name: count, dtype: int64
```

```
Attrition
No     1233
Yes     237
Name: count, dtype: int64
```

10. Correlation Matrix and Heatmap

Code:

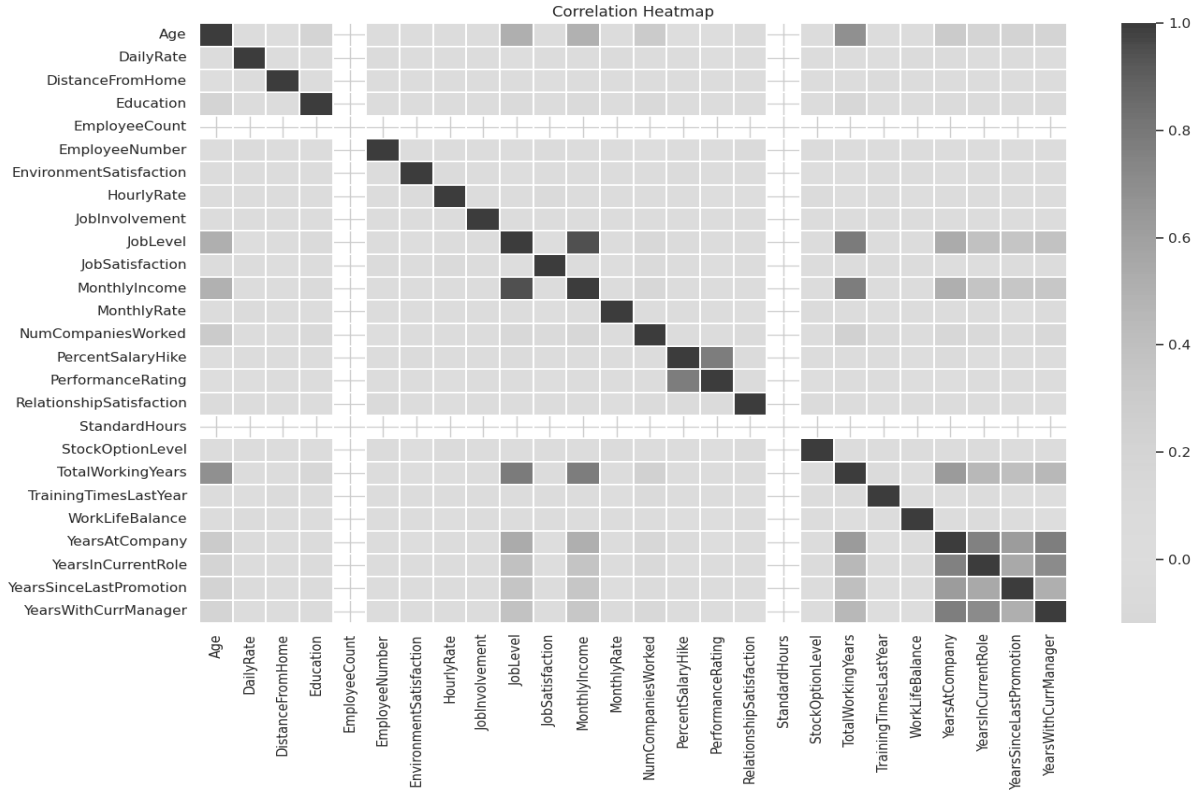
```
numeric_df = df.select_dtypes(include=np.number)

correlation_matrix = numeric_df.corr()

plt.figure(figsize=(14, 10))
sns.heatmap(
    correlation_matrix,
    cmap="coolwarm",
    center=0,
    linewidths=0.3
)

plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

```
<Figure size 1400x1000 with 2 Axes>
```



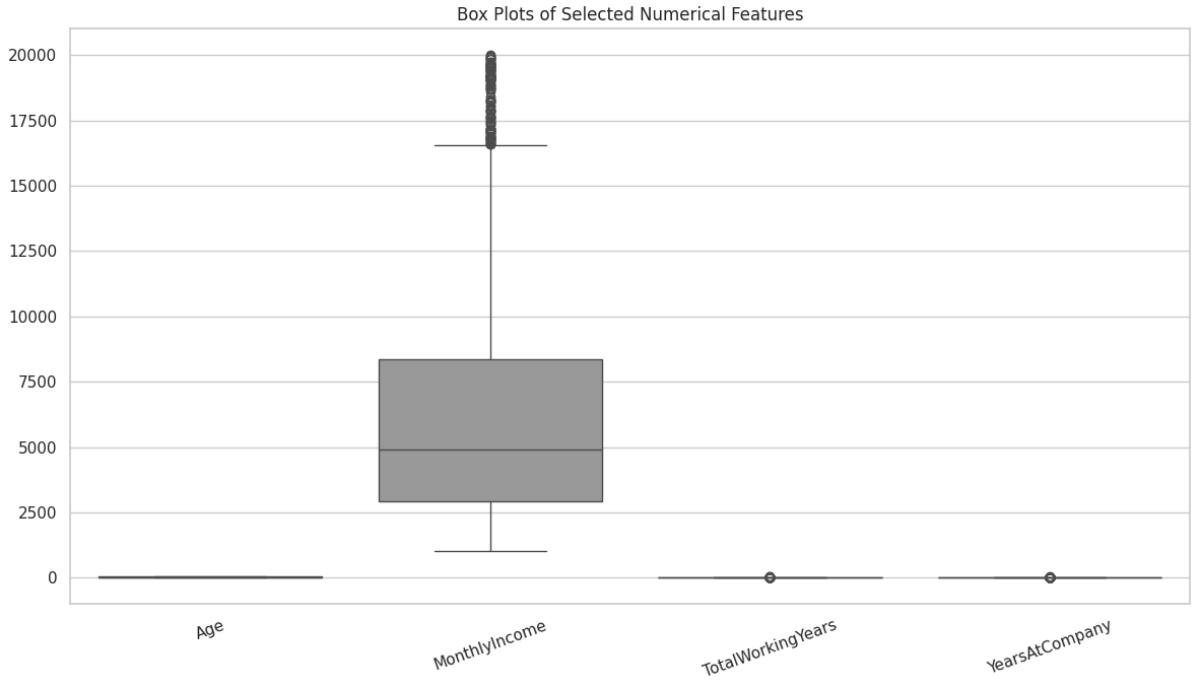
11. Box Plot for Outlier Detection

Code:

```
selected_box_columns = [
    col for col in ["Age", "MonthlyIncome", "TotalWorkingYears", "YearsAtCompany"]
    if col in df.columns
]

plt.figure(figsize=(12, 7))
sns.boxplot(data=df[selected_box_columns])
plt.title("Box Plots of Selected Numerical Features")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

```
<Figure size 1200x700 with 1 Axes>
```

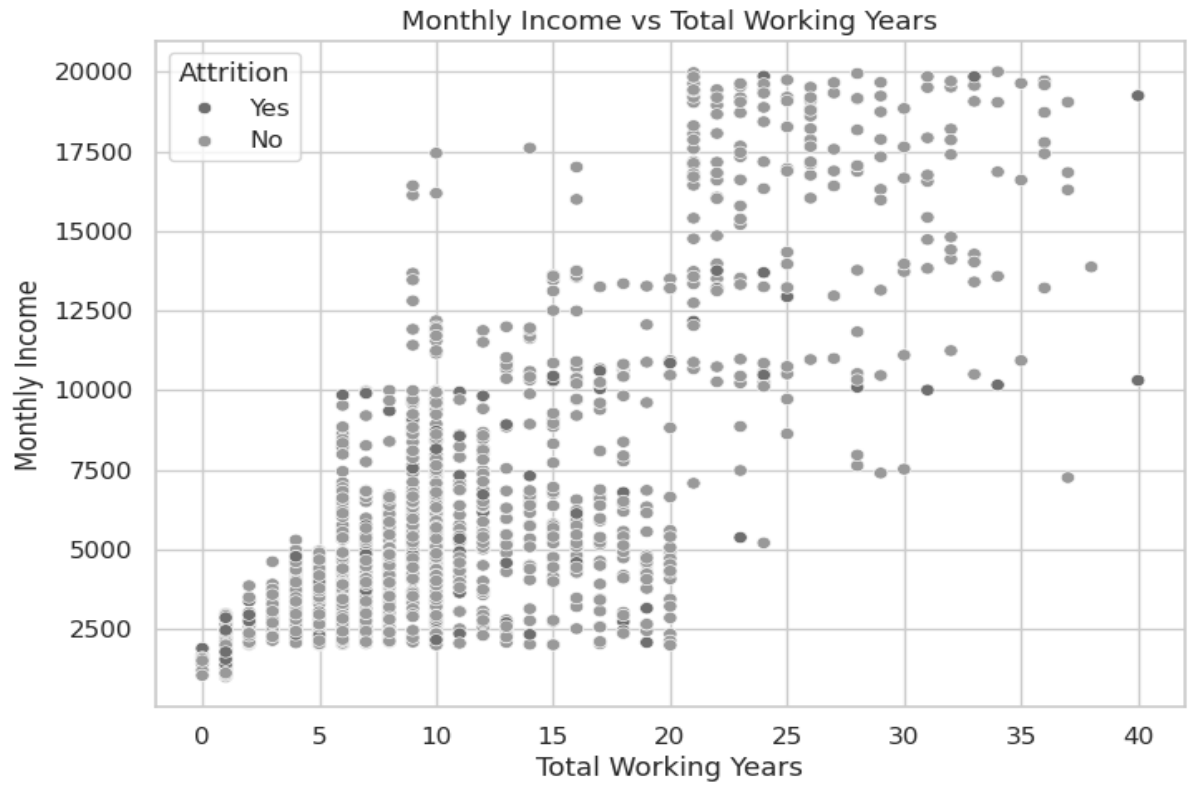


12. Scatter Plot for Feature Relationship

Code:

```
if "MonthlyIncome" in df.columns and "TotalWorkingYears" in df.columns:
    plt.figure(figsize=(8, 6))
    sns.scatterplot(
        data=df,
        x="TotalWorkingYears",
        y="MonthlyIncome",
        hue="Attrition" if "Attrition" in df.columns else None
    )
    plt.title("Monthly Income vs Total Working Years")
    plt.xlabel("Total Working Years")
    plt.ylabel("Monthly Income")
    plt.show()
```

```
<Figure size 800x600 with 1 Axes>
```



13. Univariate Analysis

Code:

```
print("Univariate Analysis")

if "Age" in df.columns:
    print("Average Age:", round(df["Age"].mean(), 2))
    print("Median Age:", round(df["Age"].median(), 2))

if "MonthlyIncome" in df.columns:
    print("Average Monthly Income:", round(df["MonthlyIncome"].mean(), 2))
    print("Median Monthly Income:", round(df["MonthlyIncome"].median(), 2))

if "Attrition" in df.columns:
    print("\nAttrition distribution:")
    display(df["Attrition"].value_counts())
```

```
Attrition
No     1233
Yes     237
Name: count, dtype: int64
```

14. Bivariate Analysis

Code:

```
if "Attrition" in df.columns and "Department" in df.columns:
    attrition_department = pd.crosstab(
        df["Department"],
        df["Attrition"],
        normalize="index"
    ) * 100

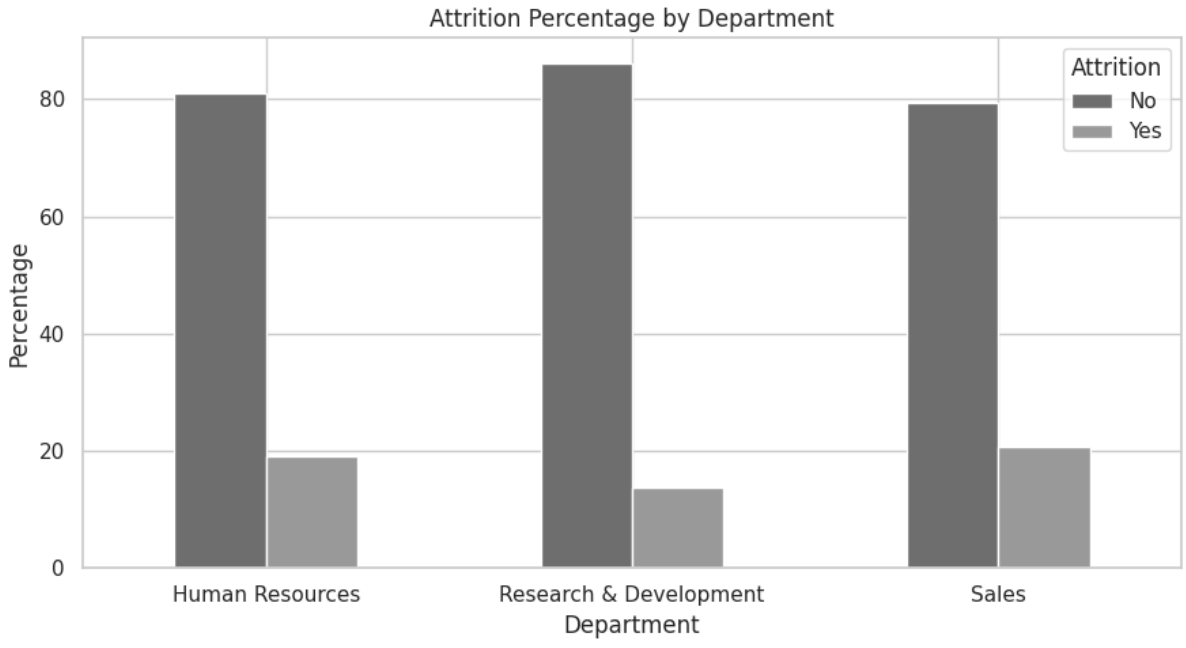
    display(attrition_department.round(2))

    attrition_department.plot(
        kind="bar",
        figsize=(9, 5)
    )

    plt.title("Attrition Percentage by Department")
    plt.ylabel("Percentage")
    plt.xticks(rotation=0)
    plt.tight_layout()
    plt.show()
```

```
Attrition                  No    Yes
Department
Human Resources         80.95  19.05
Research & Development  86.16  13.84
Sales                   79.37  20.63
```

```
<Figure size 900x500 with 1 Axes>
```



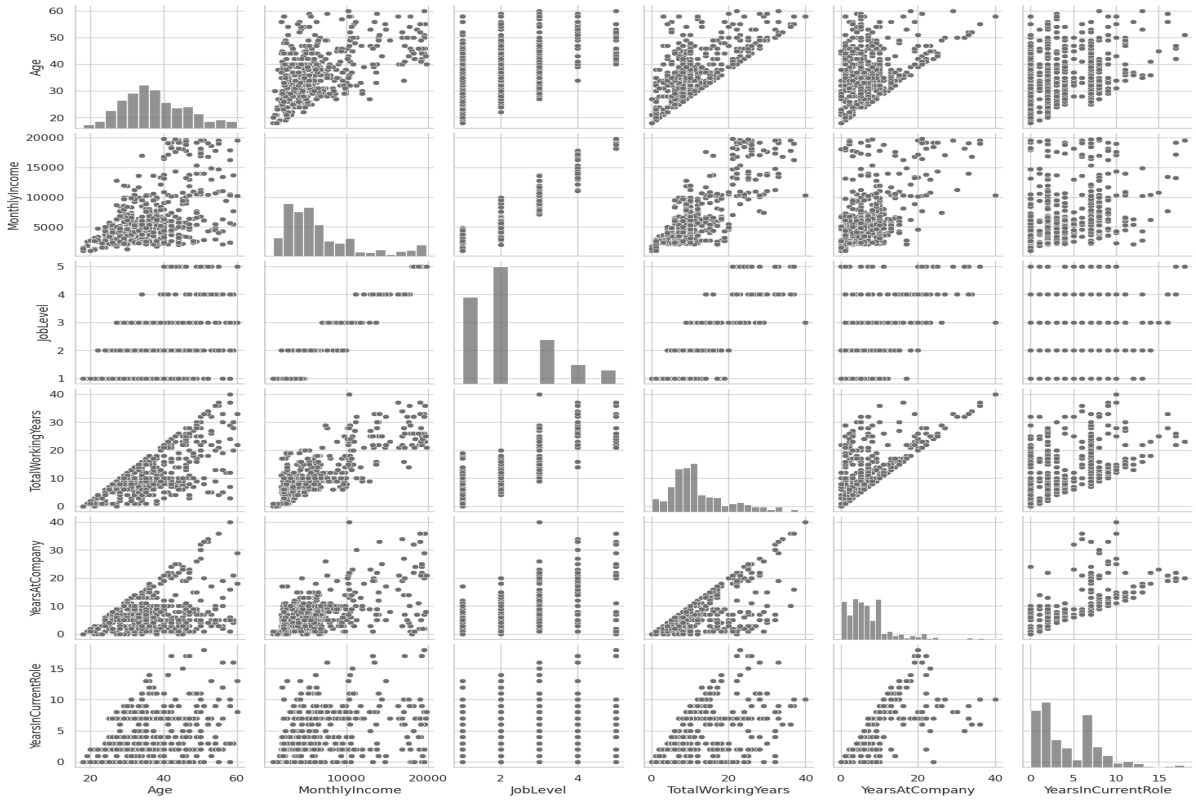
15. Multivariate Analysis

Code:

```
features = [
    col for col in [
        "Age",
        "MonthlyIncome",
        "JobLevel",
        "TotalWorkingYears",
        "YearsAtCompany",
        "YearsInCurrentRole"
    ]
    if col in df.columns
]

if len(features) >= 3:
    sns.pairplot(
        df[features].sample(min(500, len(df)), random_state=42)
    )
    plt.show()
```

```
<Figure size 1500x1500 with 42 Axes>
```



16. Identify Important Correlations

Code:

```
corr = numeric_df.corr()

pairs = []

for i in range(len(corr.columns)):
    for j in range(i + 1, len(corr.columns)):
        pairs.append((
            corr.columns[i],
            corr.columns[j],
            corr.iloc[i, j]
        ))

pairs_df = pd.DataFrame(
    pairs,
    columns=["Feature 1", "Feature 2", "Correlation"]
)

pairs_df["Absolute Correlation"] = pairs_df["Correlation"].abs()

print("Strongest feature relationships:")
display(
```

```
    pairs_df.sort_values(
        "Absolute Correlation",
        ascending=False
    ).head(10)
)
```

```
              Feature 1               Feature 2  Correlation  \
190            JobLevel           MonthlyIncome     0.950300
198            JobLevel        TotalWorkingYears     0.782208
259   PercentSalaryHike        PerformanceRating     0.773550
227       MonthlyIncome        TotalWorkingYears     0.772893
321      YearsAtCompany     YearsWithCurrManager     0.769212
319      YearsAtCompany       YearsInCurrentRole     0.758754
323  YearsInCurrentRole     YearsWithCurrManager     0.714365
18                  Age        TotalWorkingYears     0.680381
306   TotalWorkingYears           YearsAtCompany     0.628133
320      YearsAtCompany  YearsSinceLastPromotion     0.618409

     Absolute Correlation
190              0.950300
198              0.782208
259              0.773550
227              0.772893
321              0.769212
319              0.758754
323              0.714365
18               0.680381
306              0.628133
320              0.618409
```

17. Business Insights

Code:

```
print("Key EDA observations:")

if "Attrition" in df.columns:
    attrition_rate = (df["Attrition"] == "Yes").mean() * 100
    print(f"1. Overall employee attrition rate: {attrition_rate:.2f}%")

if "Department" in df.columns and "Attrition" in df.columns:
    department_attrition = (
        df.groupby("Department")["Attrition"]
        .apply(lambda x: (x == "Yes").mean() * 100)
        .sort_values(ascending=False)
    )
    print("\n2. Attrition rate by department:")
    display(department_attrition.round(2))

if "JobRole" in df.columns and "Attrition" in df.columns:
    role_attrition = (
        df.groupby("JobRole")["Attrition"]
        .apply(lambda x: (x == "Yes").mean() * 100)
        .sort_values(ascending=False)
    )
    print("\n3. Attrition rate by job role:")
    display(role_attrition.round(2))
```

```
Department
Sales                     20.63
Human Resources           19.05
Research & Development    13.84
Name: Attrition, dtype: float64
```

```
JobRole
Sales Representative         39.76
Laboratory Technician        23.94
Human Resources              23.08
Sales Executive              17.48
Research Scientist           16.10
Manufacturing Director        6.90
Healthcare Representative     6.87
Manager                       4.90
Research Director             2.50
Name: Attrition, dtype: float64
```

18. Final Observation and Recommendation

Code:

```
print("Observation:")
print("EDA helps identify distributions, relationships, anomalies, and possible factors associated with employee attrition.")

print("\nRecommendation:")
```

```
print("Business teams can use these findings to investigate departments, roles, compensation, workload, and career-growth facto
```

RESULT

Exploratory Data Analysis was successfully performed on a real-world IBM HR business dataset. The dataset structure, descriptive statistics, numerical distributions, categorical variables, correlations, outliers, and relationships between features were analyzed using Python visualizations. Business insights related to employee attrition were also identified.

Answers to Experiment Questions

Q1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

Exploratory Data Analysis is the process of examining and understanding a dataset using statistical summaries and visualizations. EDA is performed before machine learning to understand the data structure, identify missing values and outliers, detect patterns, study relationships, and discover potential data-quality problems. It helps in selecting appropriate features and models.

Q2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

Univariate analysis studies one variable at a time. Example: analyzing the distribution of employee age using a histogram.

Bivariate analysis studies the relationship between two variables. Example: analyzing Monthly Income against Total Working Years using a scatter plot.

Multivariate analysis studies three or more variables simultaneously. Example: analyzing Age, Monthly Income, Job Level, and Years at Company using a pair plot or correlation analysis.

Q3. What insights can be obtained from a correlation heatmap?

A correlation heatmap shows the strength and direction of relationships between numerical variables. Positive correlation indicates that variables tend to increase together, while negative correlation indicates an inverse relationship. It can help identify strongly related features and possible multicollinearity.

Q4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

Histograms show the distribution and spread of a numerical variable. Box plots summarize the distribution and help identify potential outliers. Scatter plots show the relationship between two numerical variables and can reveal trends, clusters, and unusual observations.

Q5. How can EDA help identify data quality issues before analysis?

EDA can reveal missing values, duplicate records, inconsistent categories, unusual values, outliers, skewed distributions, incorrect data types, and other anomalies. Detecting these issues early allows them to be addressed before statistical analysis or model building.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

Correlation is useful because it indicates how strongly numerical variables are associated and can help identify useful features. However, correlation does not imply causation. Two variables may be correlated because of another underlying factor or coincidence.

Q7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

For categorical variables, bar charts and count plots are useful because they show the frequency of each category. For numerical variables, histograms and density plots show distributions, box plots show spread and outliers, and scatter plots show relationships between numerical variables.

Q8. What business insights can be derived from the Netflix (or Superstore/HR Analytics) dataset through EDA?

For the HR Analytics dataset, EDA can reveal employee attrition patterns across departments and job roles, relationships between income and experience, distributions of employee characteristics, and possible factors associated with employee turnover. These findings can support workforce planning and retention analysis.

Q9. How does EDA contribute to feature selection and model building?

EDA helps identify relevant variables, highly correlated features, redundant information, outliers, and distributions that may affect model performance. These findings can guide feature selection, transformation, scaling, and the choice of an appropriate machine learning model.

Q10. What challenges might arise while performing EDA on large-scale real-world datasets?

Challenges include high computational and memory requirements, large numbers of variables, missing and inconsistent data, noisy observations, visualization difficulties, slow processing, and difficulty identifying meaningful patterns in very large datasets. Sampling, efficient data-processing tools, and suitable visualizations can help address these challenges.